Predicting seasonal equity offering under–pricing for optimized trading strategies.

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FINACLE
Reaching the Peak of Financial Excellence

Executive Summary

FINACLE, a financial analytics and strategy firm, presents this report to Citadel, highlighting a data-driven approach to capitalize on SEO under-pricing—a recurring market inefficiency where new shares are often priced below their immediate trading value, offering profit potential to informed investors.

Leveraging over 11,000 historical SEO observations and comprehensive macroeconomic and microeconomic data, we developed a suite of machine learning models to predict under-pricing. While initial model evaluation using R Squared indicated a need for further explanatory power, our leading model, LightGBM, demonstrated superior predictive accuracy (RMSE of 0.05277) and the ability to identify key drivers of under-pricing such as Offer Price, Trading Volume, Industry Monthly Return, and Projected Growth Rate.

Crucially, we translated these insights into a tangible trading strategy, evaluated through a robust portfolio back test. The simulation yielded compelling results: a Total Return of 22.6% and an impressive Sharpe Ratio of 1.15, indicating strong risk adjusted returns. Despite a modest Compound Annual Growth Rate (CAGR) of 5.2% and a daily Alpha of 0.03%, the strategy exhibited remarkable stability with a Maximum Drawdown of only 3.9%. This low drawdown is particularly significant for a fund like Citadel, emphasizing capital preservation and robust risk management.

While further research into features like prospectus text analysis and real time market data integration is recommended, this report demonstrates a viable, model driven approach to generating consistent alpha from SEO under-pricing. The framework presented offers strategic insights for Citadel, potentially enabling the identification of high probability SEO events for short term alpha, informing portfolio rebalancing, and supporting enhanced risk management.

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About Finacle

FINACLE is a specialized business consulting firm focused on financial analytics and strategy. With a core team of five professionals, we deliver data-driven insights tailored to the needs of leading financial institutions. Guided by our motto, *"Reaching the Peak of Financial Excellence,"* we combine analytical rigor with strategic thinking to support high-impact decision-making.

This report, **"Predicting Seasonal Equity Offering Under–Pricing for Optimized Trading Strategies,"** reflects our commitment to delivering actionable, model-driven solutions to complex market challenges.

Meet our Team

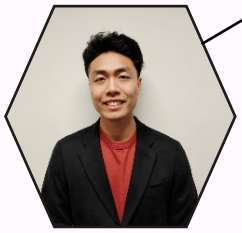
All team members served as consultants and analysts, contributing to data analysis, modelling, and strategic development.

Dahyun Ryu – Analyst

Assisted leadership and reporting.

Hanrui Zhou – Analyst

Data exploration and preprocessing.

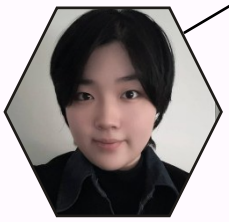


Supachai (Patrick) – Lead Data Analyst

Managed project workflow and coordination.

Manuel Fernandes – Analyst

Handled data modelling and backend coding.



Xiaomeng Mao – Analyst

Handled data preprocessing and backend coding.

1 Introduction

1.1 Background

A Seasoned Equity Offering (SEO), also known as a Follow-On Public Offering (FPO), is the issuance of new stock by a company that is already publicly traded and has previously completed an Initial Public Offering (IPO). Unlike an IPO, where a company sells its shares to the public for the first time, an SEO involves a company that already has its shares listed and trading on a stock exchange.

While raising additional equity capital is the primary outcome, companies strategically undertake **Seasoned Equity Offerings (SEOs)** for a range of critical objectives. As depicted in the 'Strategic Benefits of Seasoned Equity Offerings' figure 1.1, these include:

- **Funding Growth Initiatives:** Capital raised directly finances expansion projects like new facilities, market entry, or significant R&D.
- **Financing Acquisitions:** SEO proceeds enable external growth through mergers and acquisitions, expanding market share or capabilities.
- **Debt Reduction:** New equity can pay down existing debt, improving the balance sheet, reducing leverage, and cutting interest expenses.
- **Strengthening Financial Position:** Increasing the equity base fundamentally enhances financial health, providing a cushion against downturns and fostering stability.
- **Meeting Working Capital Needs:** SEOs provide essential liquidity for day-to-day operations, crucial for high-growth or cyclical industries.

Figure 1.1: Strategic Benefits of Seasoned Equity Offerings

However, SEOs are also associated with inherent inefficiencies that present significant opportunities for astute investors, most notably the phenomenon of under-pricing, which is the central theme of this report, and the potential for negative market reactions to the announcement of an SEO, often attributed to information asymmetry and dilution concerns.

The complexity of under-pricing, driven by macroeconomic trends (e.g., GDP growth), market sentiment (e.g., VIX), and firm metrics (e.g., P/E ratios), necessitates advanced predictive tools. This project employs a data-driven approach, leveraging machine learning to forecast under-pricing and optimize trading strategies, aligning with the needs of sophisticated investors like those at Citadel.

1.2 Objectives

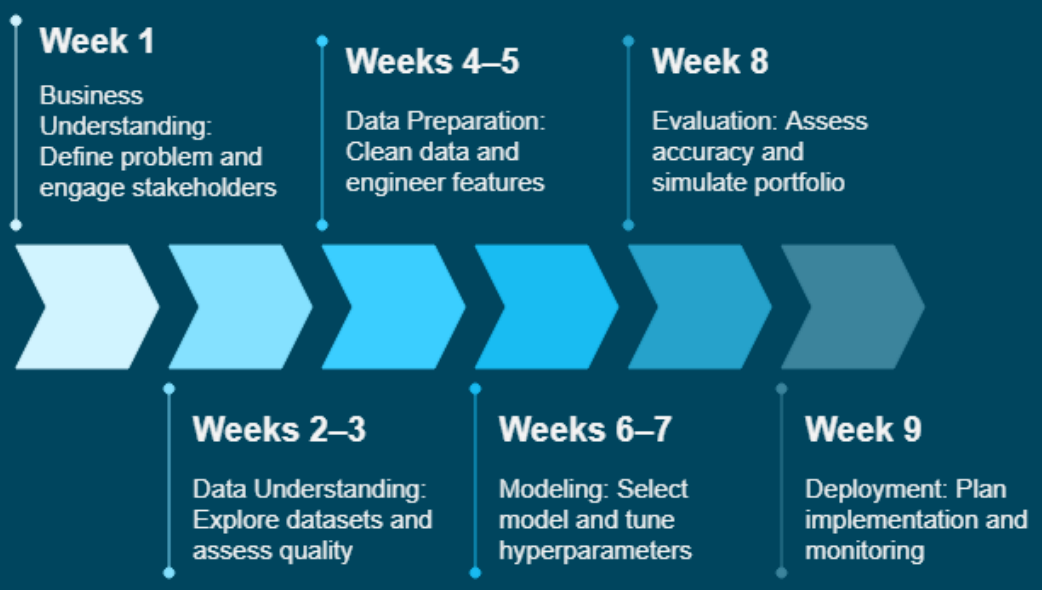
The project aims to:

- **Identify Drivers**: Uncover macroeconomic, microeconomic, and industry factors influencing SEO under-pricing.
- **Achieve High Accuracy**: Develop a model with low RMSE to ensure reliable predictions.
- **Optimize Trading**: Design a portfolio strategy that maximizes returns based on predicted under-pricing.

1.3 Project Roadmap

The CRISP-DM methodology structures the project across six phases, ensuring alignment with business goals. Figure 1.1 gives a visual view of the roadmap of our report.

Figure 1.1: Project Roadmap



2 Business Understanding

2.1 Seasoned Equity Offering (SEO) Market Context

Seasoned Equity Offering (SEO) under-pricing describes the phenomenon where new shares are offered at a price below their immediate aftermarket trading value. This represents the critical difference between the SEO's offer price (P_{offer}) and the market price (P_{market}) shortly after issuance. Specifically, underpricing occurs when $P_{market}>P_{offer}$, and its magnitude is often quantified as an initial return.

This 'money left on the table' directly costs the issuing firm, as it undercapitalizes them relative to immediate aftermarket valuations. For investors allocated shares at P_{offer} and selling at P_{market}, under-pricing presents an immediate, often substantial, profit opportunity.

2.2 Measuring SEO Under-pricing (Empirical Approach)

The most common way to measure SEO under-pricing is by calculating the **initial return.** This initial return was then converted in terms of percentages.

$$UnderPricing_{PCT} = \frac{Price\ at\ close\ issue\ date - Offer\ Price}{Offer\ Price} \times 100$$

Where:

- **Price at close issue date** is the closing price of the stock on the issue date of the SEO.
- **Offer price USD** is the offer price per share in the SEO.

Predicting under-pricing can unlock significant alpha[1] for traders, particularly in high-frequency or quantitative funds.

[1] *Alpha refers to the excess return an investment (or trading strategy) generates above a benchmark or expected return, typically adjusted for risk.*

2.3 Historical Trends and Magnitude of SEO Under-pricing

SEO under-pricing, while generally less pronounced than IPO under-pricing, is a persistent and well-documented phenomenon across global capital markets and various time periods.

Average Magnitude: Numerous academic studies consistently quantify SEO underpricing, presenting a recurring market anomaly. For US markets, average raw initial returns for SEOs have historically ranged from 1% to 5%, though this varies significantly with market conditions and offering specifics. Notably, **Kim and Shin (2001)**, examining over 3,000 US SEOs from 1983 to 1998, found a mean underpricing of 2.99% and a median of 2.02%. Their research compellingly links the increase in significant SEO discounts to the introduction of SEC Rule 10b-21, suggesting a structural driver behind greater underpricing. While these levels are considerably lower than the 15% to 20% average underpricing often seen in IPOs, they still represent a substantial cost to issuers and, critically, a predictable profit opportunity for investors.

Variability Over Time: The level of SEO under-pricing is not static. It tends to fluctuate with overall market sentiment and economic conditions.

"Hot" vs. "Cold" Issue Markets: During "hot" issue markets, characterized by high investor optimism and strong market performance, SEO under-pricing tends to be higher. Conversely, in "cold" issue markets, under-pricing may be lower.

Economic Cycles: Recessions or periods of high market volatility can also impact the level of under-pricing.

International Evidence: SEO under-pricing is not unique to the U.S. market. Studies in various other countries have also documented positive average initial returns for SEOs.

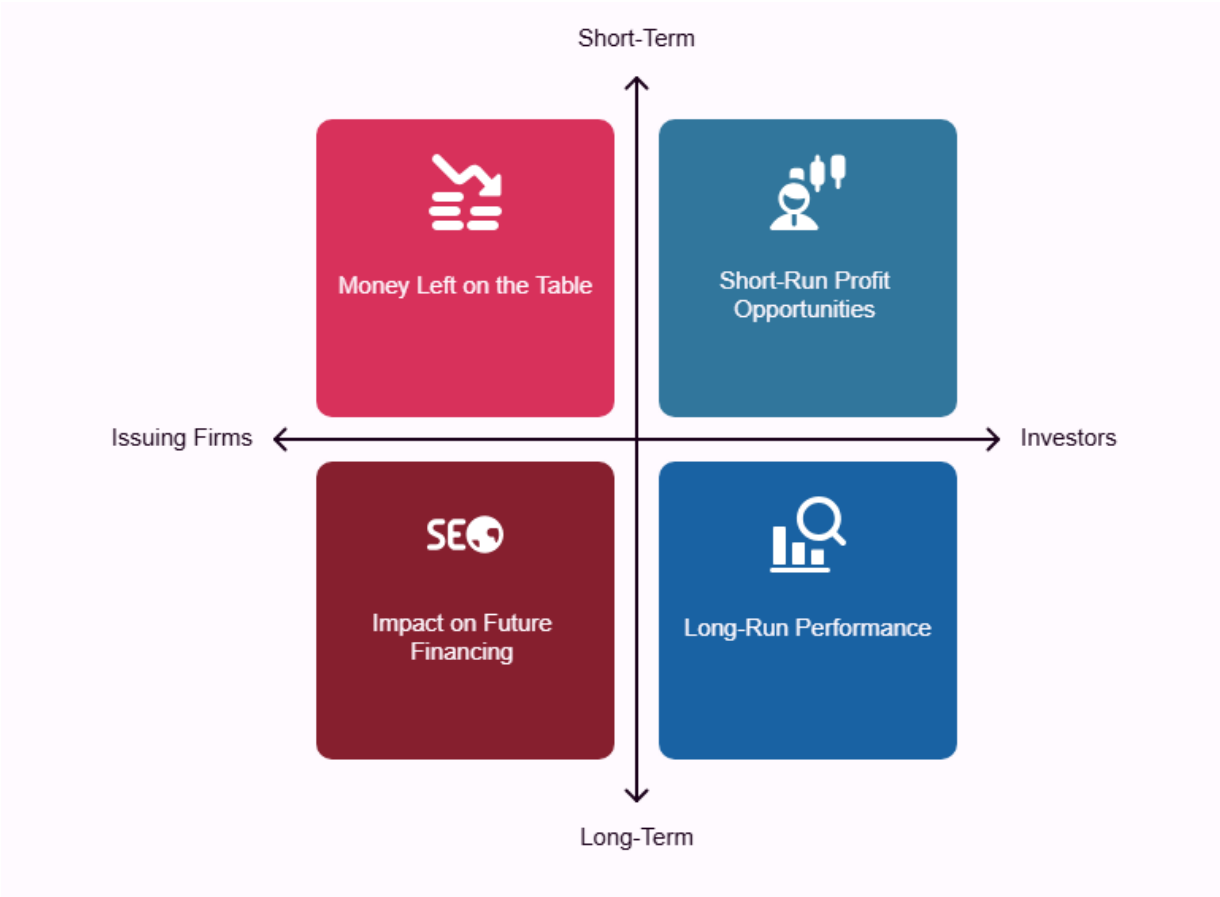
Comparison with IPO Under-pricing: SEO under-pricing is typically much lower than IPO under-pricing due to the existing market price benchmark for SEO issuers.

Recent Trends: Accelerated offering methods aim to reduce uncertainty and potentially impact under-pricing levels, though some degree often persists.

2.4 Significance for Issuing Firms and Investors

SEO under-pricing has significant economic consequences for both the companies issuing the equity and the investors participating in these offerings. For a financial institution like Citadel, understanding these significances is paramount. The ability to accurately assess and predict under-pricing, as explored in our model, can inform trading strategies, risk management, and advisory services.

Figure 2.1: Consequences of SEO Under-Pricing



For Issuing Firms:

"Money Left on the Table": The most direct consequence is the lost capital.

Dilution of Existing Shareholders' Value: Greater-than-necessary dilution occurs when shares are sold below their immediate aftermarket price.

Reputational Concerns: Consistently high under-pricing might raise questions about efficiency.

Impact on Future Financing: Terms of one SEO might influence future efforts.

For Investors:

Short-Run Profit Opportunities: The primary allure for many short-term investors.

Allocation Issues: Attractive underpriced SEOs are often oversubscribed.

Long-Run Performance: Many studies find that firms tend to underperform market benchmarks in the years following an SEO.

Information Content: The degree of under-pricing can be interpreted as a signal by sophisticated investors

Methodology & Analysis

3.1 Data Sources

[2]The analysis leverages datasets categorised into four groups:

Raw SEO Dataset: 14,308 observations, 43 variables (e.g., Offer Price in USD, Issue Price, CUSIP9), provided by Client (data from 2010-2023)

Industry Portfolio Data: Fama-French factors (e.g., Mkt-RF, SMB) and value-weighted returns.

Macroeconomic Data: 25 variables (e.g., personal consumption, GDP percentage change, consumer price index).

Microeconomic Data: 11 firm metrics (e.g., price-to-earnings ratio, net profit margin).

A comprehensive dataset was constructed by sourcing and integrating data from multiple providers. The primary dataset contained details on individual SEOs like issuer information, offer price, shares offered, issue type, exchange, and post-issue prices (*Price_at_Close_Issue_Date, Price_1_Day_After*, etc.). Mapping of *CUSIPs*[2] to *CIKs*[2] was conducted using SEC EDGAR API to retrieve company-specific details such as state in which the company is headquartered. To fill in missing issue prices, daily security data was sourced from Wharton Research Data Services (WRDS) using a combination of CUSIPs and Ticker values.

Macroeconomic data was sourced from Bureau of Economic Analysis (BEA), Wharton Research Data Services (WRDS) and St. Louis Federal Reserve Economic Data. Market Data and Returns data were collected from Eugene F. Fama and Kenneth R. French. For considering market sentiments in our model CBOE Volatility Index closing prices were considered. Lastly Financial ratios for SEOs

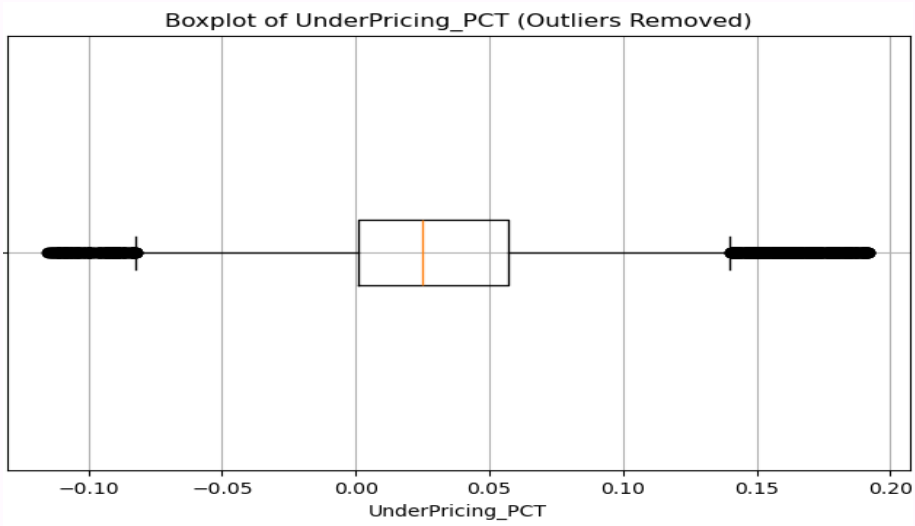
[2] ***For variable definitions, please refer to Appendix A***

were sourced from Compustat – WRDS. These diverse datasets merged in to create a rich analytical dataset.

3.2 Insights from Exploratory data analysis (EDA)

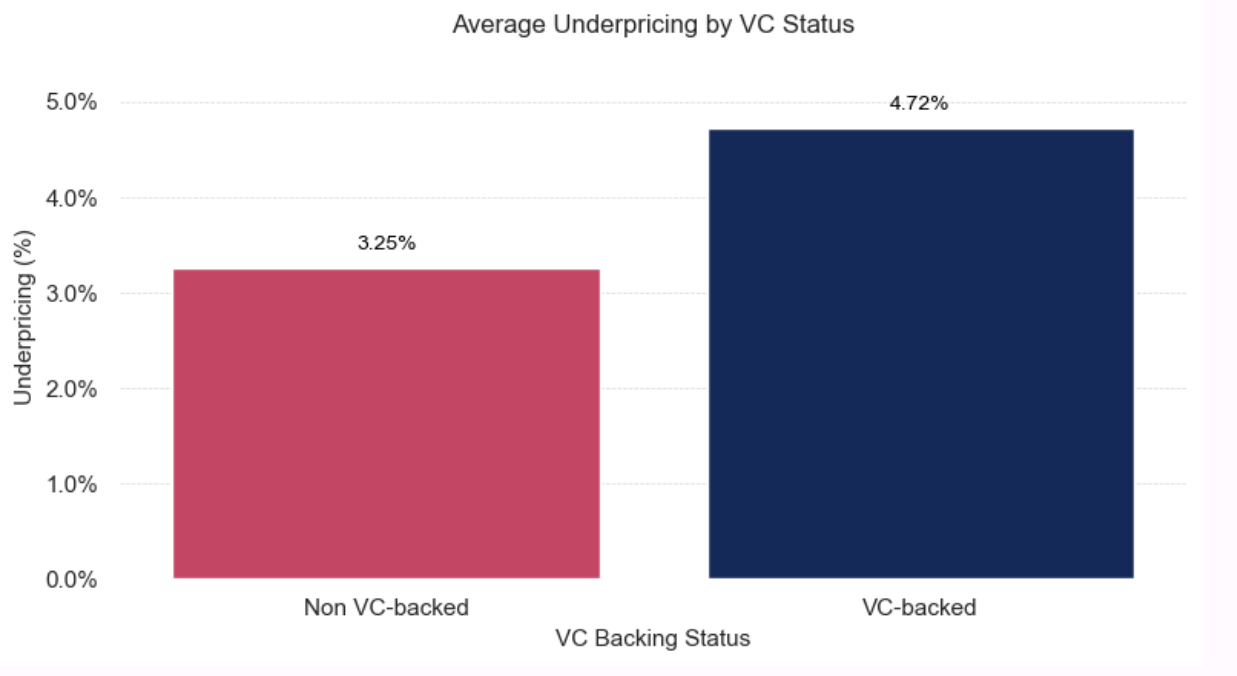
The Original Dataset had **14,308** observations, after formatting and filtering, the total number of observations for EDA stood at **11,797**. The distribution of Under-pricing was examined, and outliers were handled using interquartile range (IQR) method, suggesting that extreme under-pricing events exist and might skew analyses if not treated.

Figure 3.1: Boxplot showing data range of target variable



The boxplot illustrates the distribution of under-pricing percentages for the SEOs, with outliers removed. The central box represents the IQR, showing that the middle 50% of observations fall between approximately 0.00 and 0.05. The median under-pricing percentage is around 0.02 to 0.03, indicating a slight **positive** under-pricing tendency. The whiskers extend to approximately -0.08 and 0.13, representing the range of the remaining data, thus showing the total range of under-pricing. **In summary**, while under-pricing varies somewhat, a typical SEO in this dataset experiences modest positive under-pricing which can derive profits through investing.

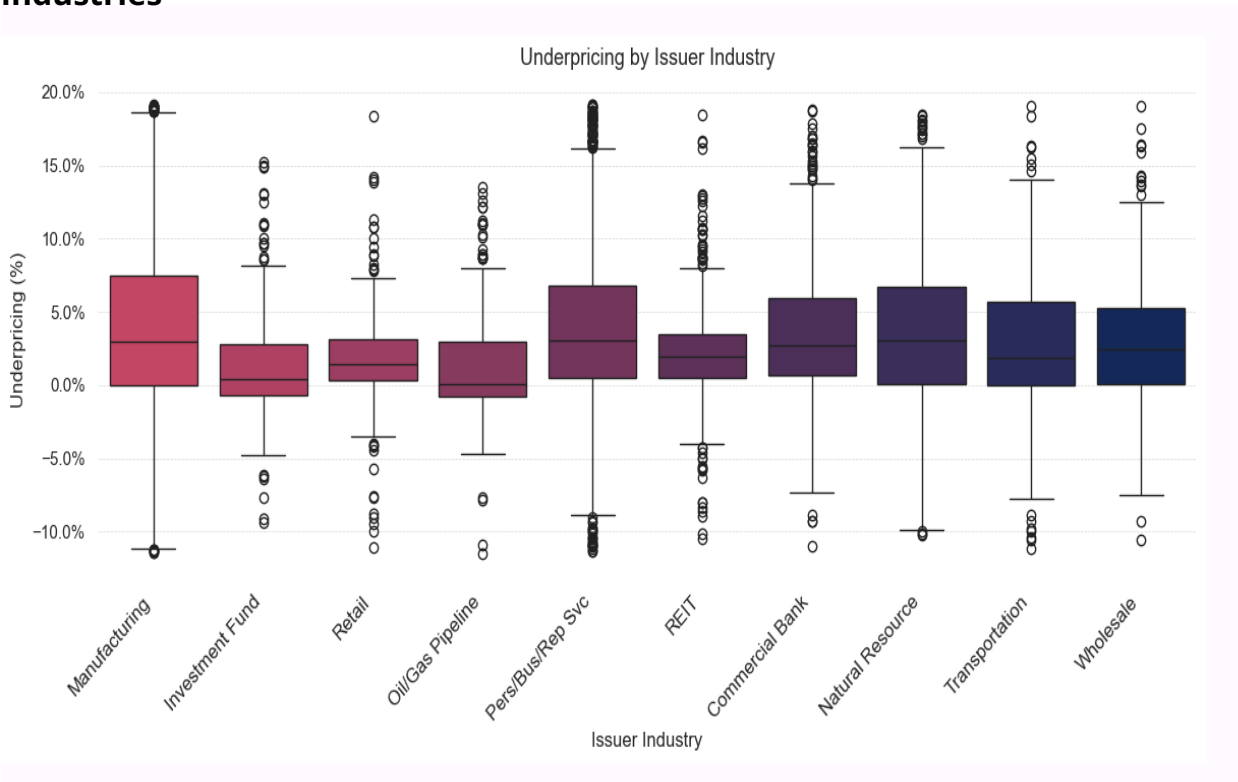
Figure 3.2: Average under-pricing percentage by Venture Capitalist (VC) backed SEOs



The bar chart compares the average under-pricing percentage (*UnderPricing_PCT*) for SEOs, broken down by VC-backed status. SEOs of VC-backed firms exhibit a higher average under-pricing of 4.72%, compared to non-VC-backed firms, which have an average under-pricing of 3.25%. This **suggests** that SEOs issued by companies with venture capital backing tend to be more underpriced than those without VC backing.

Upon further review, we found some more interesting patterns which helped in refining the dataset before modelling.

Figure 3.3: Boxplot showing data range of underpricing across different industries



The boxplot chart compares the distribution of under-pricing percentages across different issuer industries. It reveals that under-pricing varies considerably across industries. For example, some industries, such as 'Manufacturing' and 'Commercial Bank', exhibit a wider range of under-pricing percentages, including both higher and lower outliers, while others, like 'Oil/Gas Pipeline', show less variability. The median under-pricing also differs across industries, **suggesting that the typical under-pricing level is industry specific**. Overall, the chart highlights that an issuer's industry plays a significant role in determining the extent and variability of under-pricing.

Figure 3.4: Under-pricing percentage by US States

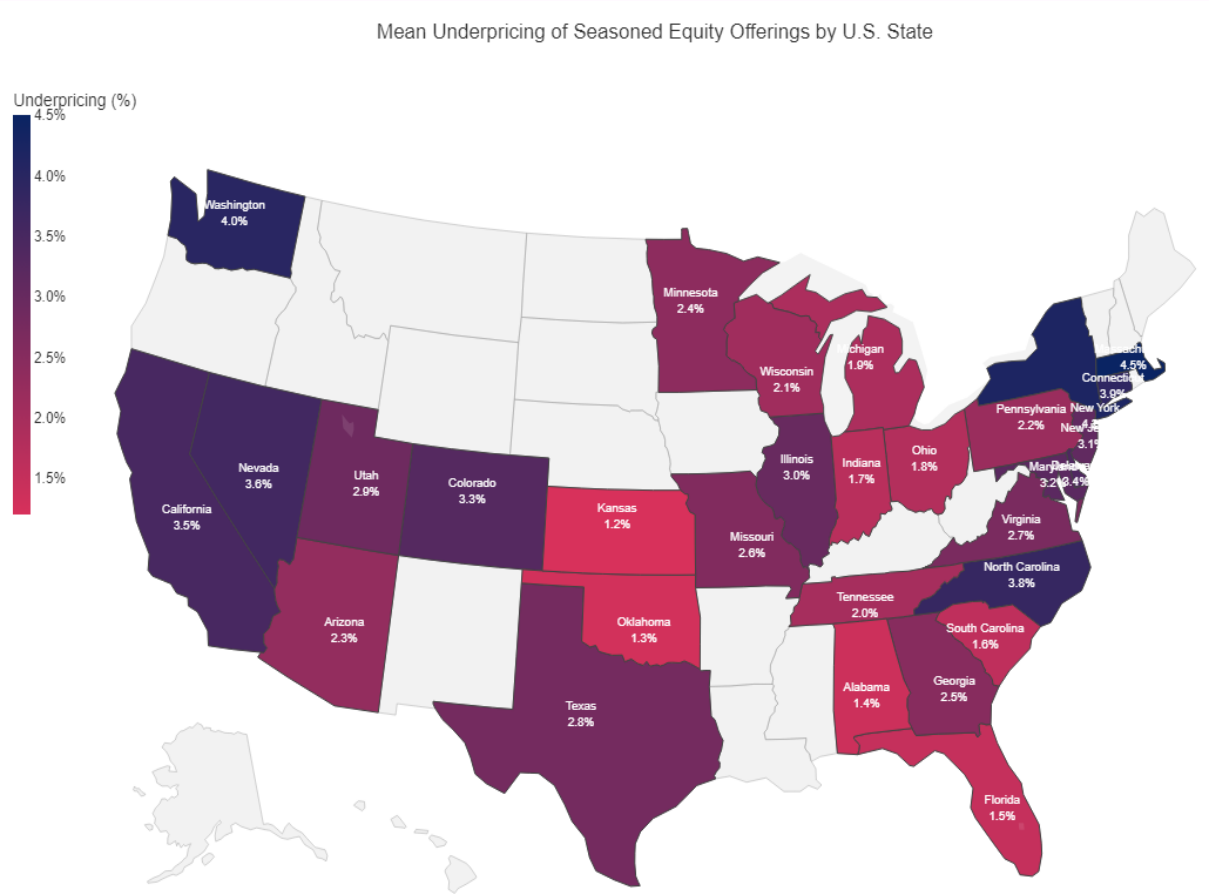


Figure 3.4 demonstrates that the average under-pricing percentage of seasoned equity offerings (SEOs) is notably higher in states such as Minnesota, Utah, and Nevada, when compared to states like Illinois and Virginia. This geographical disparity suggests that firms headquartered in certain regions are more susceptible to SEO under-pricing. Several factors could contribute to this variation, including differences in investor sentiment, local market dynamics, regulatory environments, or firm-specific characteristics prevalent in those states.

From an issuer's perspective, under-pricing represents a potential cost, as it implies that shares are sold below their intrinsic or fair market value, resulting in less capital raised than could have been achieved under more favourable pricing conditions.

Conversely, from the investors' standpoint, under-pricing may be perceived as an opportunity, allowing them to acquire shares at a discount relative to their post-offering market value. Therefore, the implications of under-pricing are inherently perspective-dependent, reflecting a trade-off between issuer capital efficiency and investor return potential.

3.3 Data Preprocessing and Feature Engineering

Significant preprocessing steps were undertaken:

- **Target variable calculation:** Under-pricing percentage was calculated using the formula mentioned earlier under Business Understanding.
- **Handling missing observations:** As mentioned previously, missing observations for critical variables such as the closing issue price and firm identifiers were supplemented using recognized external data sources. This imputation was performed only after a rigorous verification process to minimize the risk of introducing inaccuracies. Variables with more than 20% missing values were excluded from the analysis, as imputing such a significant proportion would compromise both the robustness of the model and computational efficiency. For macro-economic indicators expressed as percentage changes, missing values were imputed with zero, under the assumption of no change during the corresponding period.
- **Outlier Treatment:** Extreme values from *under-pricing percentage* were excluded from analyses to enhance the robustness of results and reduce the influence of skewed data distributions.
- **Log Transformations:** We created a new variable, **log_trading_vol**, to smooth out extreme trading volumes and reveal clearer patterns. Zero volumes were treated as minimal activity to avoid errors. This helps us better assess how investor interest relates to under-pricing.

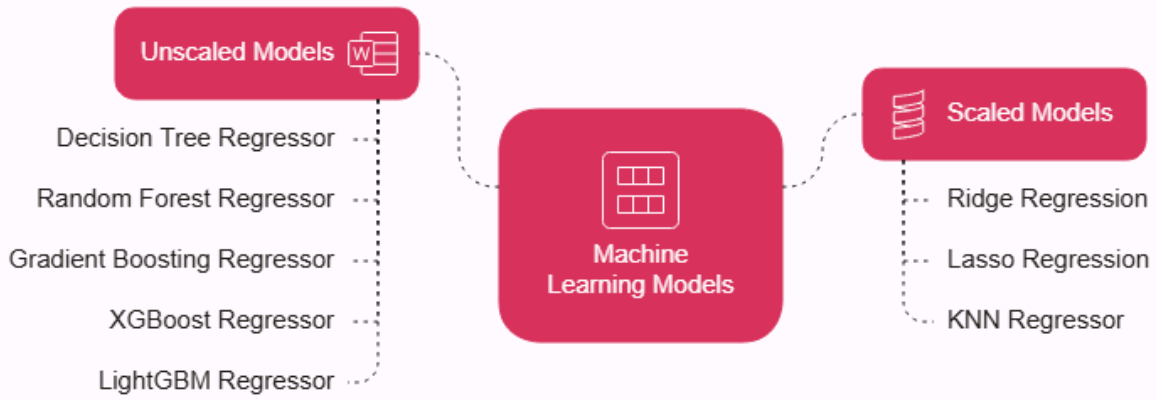
- **Categorical feature encoding:** We converted text-based details like stock exchange, issue type, and industry into a format that models can understand. Each category became its own column, except one from each group, which we left out to keep the analysis clean and avoid overlap.
- **Date features:** From the *Issue Date* and *Offer Date*, we created new details like year, month, and quarter of the offering. Most importantly, dates were **lagged before merging** with external data (like market conditions), to ensure we only used information that would have been known **before** the offering. This avoids accidentally using future data and makes the analysis more realistic and accurate.
- **Removing miscellaneous columns:** Columns that didn't add useful information, such as ID numbers or those with the same value in nearly every row (like *IPO_Flag* column), were removed. This helps keep the data clean and focused, making the analysis more efficient and meaningful.

4. Predictive SEO Modelling

At the beginning of our analysis, the dataset contained **11,797** observations. However, after performing several data cleaning and processing steps, such as handling missing values, removing outliers, and filtering out incomplete records, the number of usable observations for modelling stood at **3,829**. These steps were essential to ensure the quality and reliability of the final dataset used for building the model.

A comprehensive suite of machine learning models (mentioned *in figure 4.1*) were developed to predict the *Under-Pricing Percentage*. Both scaled and unscaled models were included in the analysis to evaluate the impact of data normalization on model performance.

Figure 4.1: Model Scaling Strategies



4.1. Why Machine Learning Model?

Traditional financial modelling, such as linear regression, often relies on the assumption of a static, linear relationship between market variables (e.g., a direct, proportional impact of trading volume on under-pricing). However, the inherent complexity and dynamism of financial markets rarely conform to such simplified assumptions.

To overcome these limitations and gain a more nuanced understanding of market behaviour, we leveraged advanced machine learning techniques. Machine learning excels at identifying and modelling intricate, non-linear interdependencies—for instance, how a confluence of industry trends, market timing, and shifting investor sentiment collectively influences under-pricing.

4.2. Model Validation and Performance and Results

Time-Series Cross-Validation: We used a technique that trains the model on historical data and tests it on future data. This approach respects the natural flow of time and ensures that the model doesn't "peek" at future results, which would otherwise lead to misleading performance. This is crucial in financial modelling, where real-world decisions are always made without knowledge of future outcomes.

Hyperparameter Tuning: To improve accuracy, each model was fine-tuned using a systematic process that tests various parameter combinations. This step helps the model learn more effectively from the data and minimizes forecasting errors. The goal is to find the version of each model that performs best on unseen data

Performance Metrics: We assessed how well each model predicted under-pricing using metrics that matter in a financial context:

- **Root Mean Squared Error (RMSE)** shows how far predictions are from actual values. (Lower *RMSE = Better Model*)
- **R-squared (R^2)** indicates how much of the variation in under-pricing the model can explain. These metrics provide a clear picture of how useful each model is for decision-making. *(Higher value = Better explainability of model*)

Table 1: Model Performance Summary

MODELS	RMSE SCORES	R SQUARED
RIDGE	0.0532	-0.0122
LASSO	0.0531	-0.0068
KNN	0.0542	-0.0510
DECISION TREE	0.0541	-0.0452
RANDOM FOREST	0.05278	0.0103
GRADIENT BOOSTING	0.0530	-0.0031
XGBOOST	0.05272	0.0073
LIGHTGBM	0.05277	0.0053

From Table 1, we see that, among all the models, **LightGBM** performed very well, with one of the lowest RMSE values (0.05277) and a positive R^2 score (0.0053), meaning it slightly outperformed baseline predictions. **Random Forest** also gave reliable results, showing a positive R^2 (0.0103), indicating a decent ability to capture patterns in the data.

4.2 Why we chose LightGBM over Random Forest?

- ✓ **Superior Predictive Accuracy** - LightGBM's gradient boosting framework captures **non-linear and subtle market patterns**, delivering more accurate forecasts of price movements, investor reactions, or post-SEO performance. In our testing, it showed **lower RMSE (0.05277)** than Random Forest, making it more reliable for high-stakes decisions.
- ✓ **Speed at Scale** - LightGBM trains significantly faster, enabling **real-time scenario modelling**, essential for investors needing quick signals around SEO announcements.

- ✓ **Greater Signal Extraction** - It handles **high-dimensional financial datasets** (e.g., macro indicators, liquidity ratios, sentiment scores) with ease, revealing alpha-generating insights that Random Forest might miss due to its ensemble averaging nature.
- ✓ **Deployment-Ready for Live Trading Models** - Its performance and scalability make LightGBM well-suited for integration into **automated investment strategies and risk models**.

4.3. Identifying Key Predictors from Models

The LightGBM feature importances reveal key drivers of SEO under-pricing. For an institution like Citadel, understanding these factors helps refine predictive models, guide investment strategies, and advise clients on SEO structuring.

Figure 4.2: Top 10 Variables provided by LightGBM

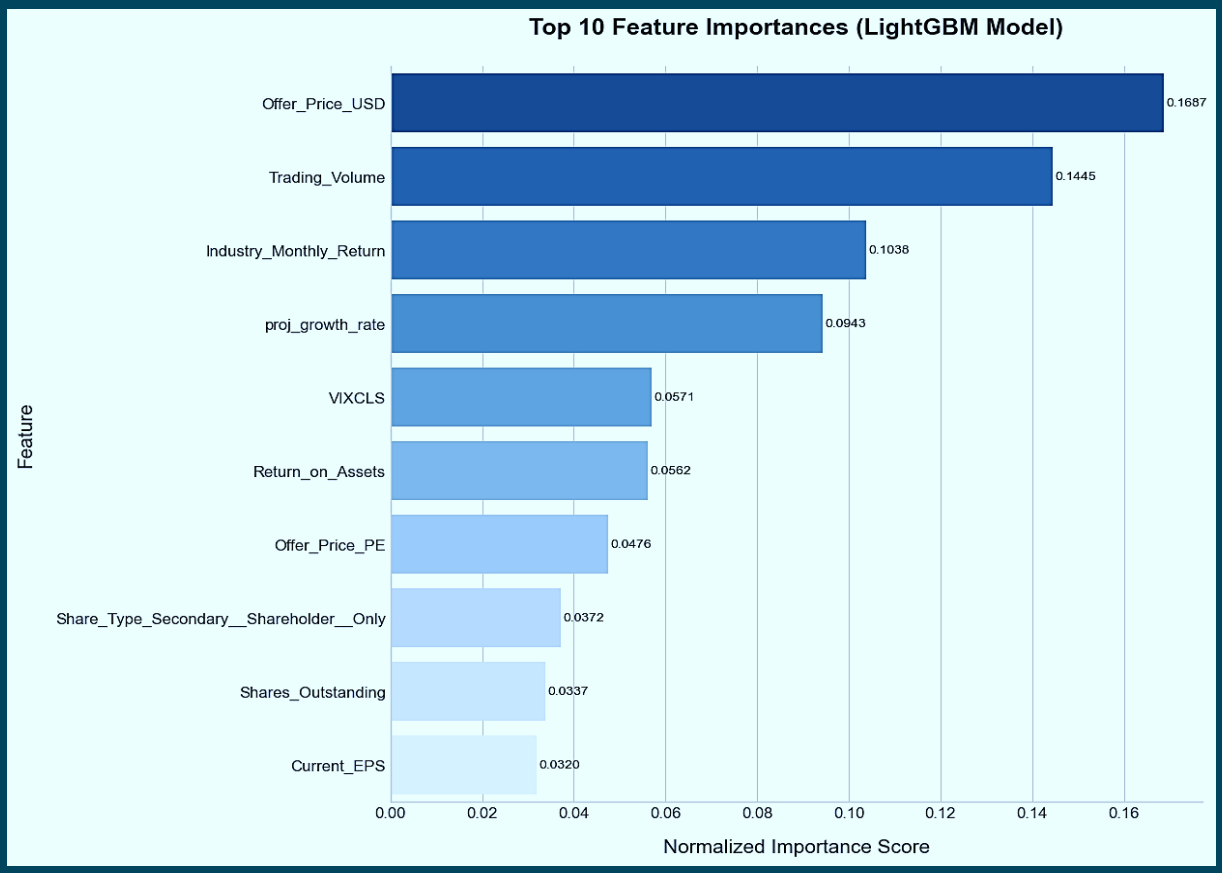


Figure 4.2 shows the Offer Price (USD) as the most influential factor, driving investor sentiment; a mispriced offer can signal overvaluation and trigger negative market reactions. Beyond pricing, several factors shape investor response: robust Trading Volume suggests institutional interest, often cushioning price drops. Industry Monthly Return indicates that offerings in strong-performing sectors attract capital, emphasizing timing and sector alignment. Companies with strong Projected Growth Rates face less dilution backlash as investors bet on future value. Conversely, Market Volatility (*VIXCLS*) increases risk premiums, leading to muted or adverse investor responses.

Operational efficiency, reflected by Return on Assets (ROA), signals financial health and builds buy-side confidence. Offer Price-to-Earnings (P/E) aids analysts in fair pricing, impacting participation. Finally, Ownership Type, Share Count, and weak EPS can dilute existing value and depress demand.

5. Trading Strategy

5.1 Signal Generation

The model results are valuable for crafting an investment strategy focused on maximizing profits. For a financial institution like Citadel, the goal of developing predictive models for SEO under-pricing is to inform actionable trading strategies.

The trading strategy relies on predictions driven by key features. The LightGBM model captures complex interactions between variables such as *Offer_Price*, *Trading_Volume*, *Industry_Monthly_Return*, *proj_growth_rate*, and Volatility (*VIXCLS)*, forming the basis for generating trade signals.

5.2. Trading Strategy Simulation Framework

To evaluate the LightGBM model's practical applicability, a trading strategy simulation was conducted on the following assumptions:

- **Signal Generation:** An *Expected Daily Return* was calculated based on predicted under-pricing and a predefined *Holding Days* (e.g., one day, using *Price at Close Issue Date*). A *Trade Flag* was generated if the *Expected Daily Return* exceeded the threshold (e.g., 4.5% [3]).
- **Base Capital and Max Investment Limit:** An initial capital (e.g., $1,000,000) was defined. For each flagged SEO, the investment amount was determined, subject to a *Max Investment* cap (e.g., 10% [4] of the total market value of the SEO) and available capital.
- **Portfolio Management and Return Calculation:** Capital committed to a trade, cannot be reused until exit. Capital is deployed sequentially, with up to 5% of the portfolio allocated per SEO, following the principle of diversification and avoiding concentration risk by not placing all capital in a single investment.

[3] *The 4.5% and 10% threshold can be modified as per Citadel's risk appetite. However, we recommend considering 4.5% based on median under-pricing percentage of our dataset outlined in EDA*

After trade exit, the capital is added back for reinvestment.

- **Costs** – 5% of transaction value was considered as Brokerage fees, slippage and other costs.

Figure: 5.1: Strategic assumptions of Investment Strategy.

Based on the above assumptions, our trading strategy was implemented. We used a sophisticated method called "**Portfolio Back test**" (Schumann, 2018) to test our strategy for success. A portfolio back test is a method used to evaluate how a trading strategy or investment portfolio would have performed in the past using historical data. It's like asking, *"If we had followed this strategy in the past, how much money would we have made (or lost)?"*

5.3. Interpretation of Strategic Outcomes

The output of this simulation, such as the final value of our capital would reveal whether the strategy of selectively investing in SEOs based on the model's predictions and defined rules resulted in a positive return on the initial capital.

Figure 5.2: Portfolio Capital Growth over time (Greedy Strategy)

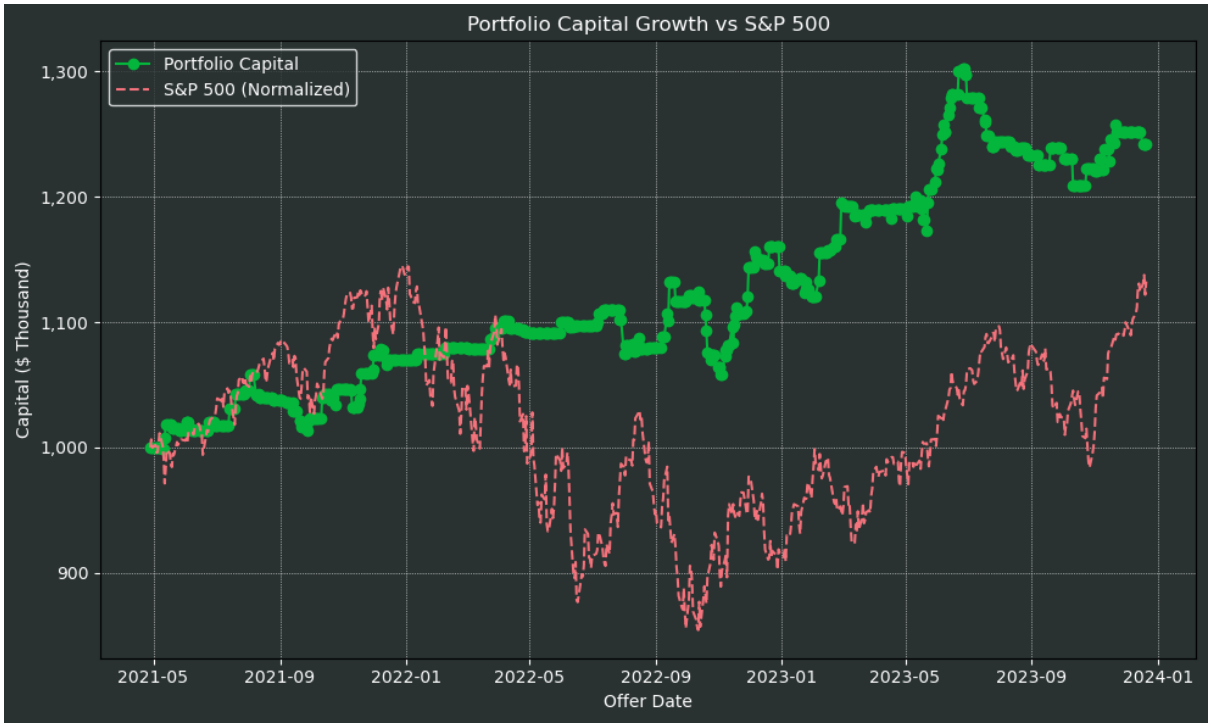


Figure 5.2. shows the simulated growth of an initial investment portfolio over time, based on the "Greedy Strategy" from the LightGBM model's SEO under-pricing predictions. The x-axis represents the Offer Date of SEOs from May 2021 to early 2024, while the y-axis shows portfolio capital in dollars, reaching up to $1.3 million. Each point reflects the portfolio's total capital on a given day, with the line connecting these points to show the capital growth trajectory. The simulation spans about two and a half years (May 2021 - early 2024), capturing various market phases.

Insights from Trading Strategy:

- **Positive Validation:** Figure 5.2 supports the efficacy of the SEO underpricing prediction model and trading strategy, suggesting the strategy successfully captured alpha.
- **Compounding Effect:** The accelerating growth in later periods indicates successful profit reinvestment.
- **Strategy Robustness:** The consistent upward trend suggests that the strategy's success was based on multiple profitable opportunities, not isolated trades.

5.4. Strategy performance evaluation

Table 2: Strategy evaluation metrics

Metric	Value	Description
Total Return	22.6%	Cumulative gain from the beginning of trade.
CAGR	5.2%	Annualized growth rate
Sharpe Ratio	1.15	Excess return over the risk-free rate
Maximum drawdown	-3.9%	The steepest peak-to-trough decline
Alpha	0.03%	Averafe daily outperformance relative to the S&P 500

Total Return: This is the overall gain the strategy would have delivered across the backtest period. For a firm like Citadel, it reflects a strong opportunity to generate profits from repeated patterns in SEO underpricing.

CAGR (Compound Annual Growth Rate): This shows the average yearly return. It may appear modest but signals a steady and repeatable edge in capturing pricing inefficiencies, important for consistent long-term performance.

Sharpe Ratio: This measures how effectively the strategy converts risk into return. A ratio above 1 means the return is not only positive but also efficiently earned. It reflects smart risk-taking and disciplined execution.

Maximum Drawdown: This is the largest decline from peak to trough in portfolio value. A low drawdown indicates strong capital preservation, giving Citadel flexibility and stability even during challenging periods.

Alpha: This is the average daily outperformance versus the S&P 500. It represents true added value from the strategy itself, independent of market trends, and supports portfolio diversification and excess return generation.

6. Conclusions and Future Directions

6.1. Hypothetical Application: Strategic insights for Citadel

A financial institution like **Citadel** could apply these analytical insights in several strategic ways:

1. Live Data Ingestion & Feature Engineering Pipeline

Data Sources:

- **Market Feeds**: Real-time trading volume, price action, and volatility (e.g., Bloomberg, Refinitiv).
- **Regulatory Filings**: Offer terms and financial metrics via EDGAR API (e.g., 424B, 8-K).
- **Industry Benchmarks**: Peer returns and sector indices for relative valuation.

Streaming Architecture:

- Stream data via **Apache Kafka** or **AWS Kinesis**.
- Trigger ETL jobs at 15-minute to 1-hour intervals, based on latency needs of each desk.
- Normalize features and store in a real-time feature store (e.g., **Tecton**, **Feast**).

2. Scoring & Alpha Signal Generation

- Deploy **LightGBM model** as an API microservice (FastAPI/Flask + Docker).
- Score new SEOs in real-time, assigning:
    - **Underpricing probabilities**
    - **Expected short-term return deltas**

- Generate ranked signals daily for tactical opportunity assessment.

3. Portfolio Execution & Rebalancing

Proprietary Trading:

- Identify high-probability SEO events for **intraday or short-term alpha**.
- Rebalance trading positions **daily or weekly** based on event windows and liquidity.

Asset Management:

- Integrate SEO scores into **multi-factor risk-return models**.
- Rebalance portfolios **monthly or quarterly**, adjusting for score volatility and market outlook.

4. Monitoring, Drift Detection & Risk Controls

Model Monitoring:

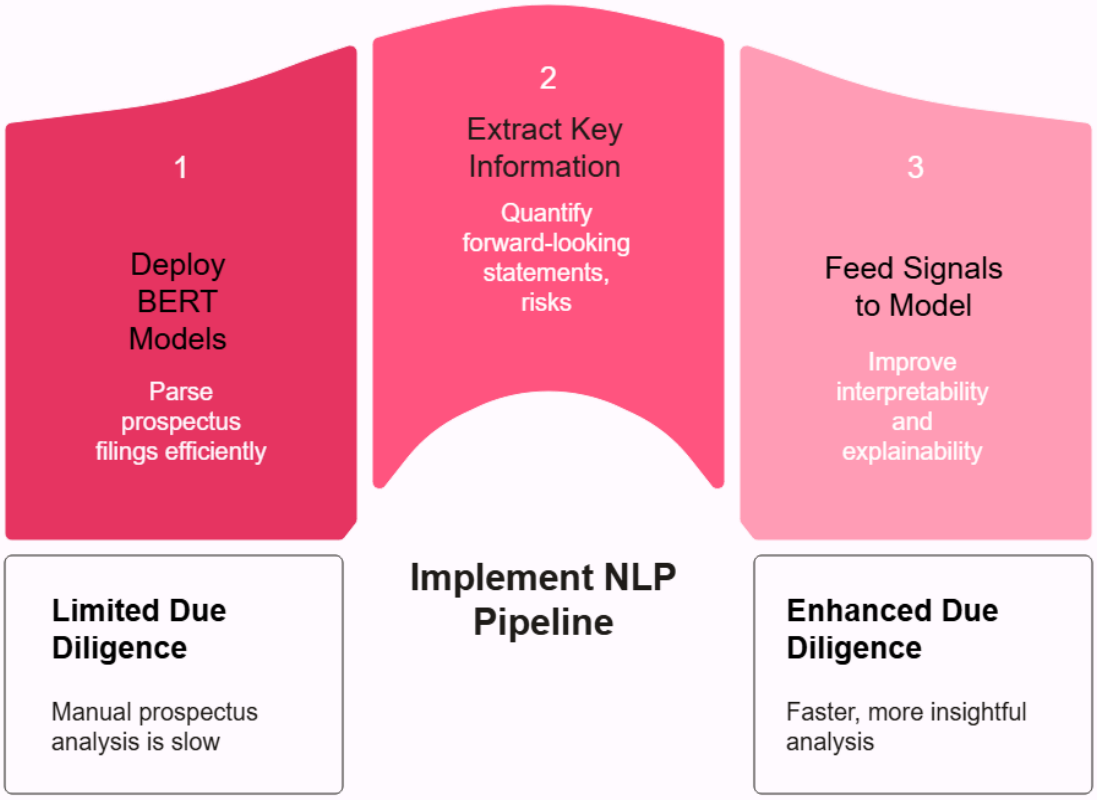
- Track feature importance, prediction drift, and data anomalies.
- Trigger retraining every **3 months**, or earlier if drift thresholds are exceeded.

Scenario Stress Testing:

- Simulate portfolio exposure under stress conditions:
    - VIX spikes
    - Earnings surprises
    - Industry shocks
- Use **Monte Carlo** or historical simulations to validate model robustness.

5. NLP Pipeline (Future Scope for Enhanced Due Diligence)

- Deploy **BERT-based NLP models** to parse prospectus filings (e.g., 8-K, 424B).
- Extract and quantify:
  - **Forward-looking statements**
  - **Risk disclosures**
  - **Sentiment tones** from management commentary
- Feed extracted signals as secondary features into the model for greater **interpretability and explainability**.



6.3. Limitations of Current Analysis (Data Quality, Model Scope)

Data Quality: Issues like missing data and constant values for some variables necessitated preprocessing that might affect generalizability.

Model Performance: While the strategy was profitable, the R2 of the predictive model was modest, indicating substantial unexplained variance.

Feature Scope: Exclusion of detailed prospectus text, news sentiment, or granular investor demand signals might limit predictive power.

Dynamic Markets: Model relationships can change; continuous monitoring and retraining are vital.

Trading Simulation Assumptions: The simulation did not include transaction costs, slippage, or detailed liquidity constraints, which would impact real-world returns.

In conclusion, this empirically grounded analysis of SEO underpricing demonstrates the potential of applying advanced business analytics and machine learning to navigate and potentially capitalize on this complex market phenomenon. The detailed data work and modeling pipeline provide a strong foundation for further refinement and strategic application by institutions like Citadel.

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APPENDIX – A: Glossary of Terms

Table 1: Variable Definitions of raw dataset (14,308 Observations)

Variable Name	Definition	Missing Values (% 36oolean)
IPO_flag	The **IPO_flag = false** means that the stocks in question have already undergone their IPO and vice-versa	**0**
original_IPO_flag	Same as "original_IPO_flag"	**0**
Price_Date_Diff	Days between closing and issue dates.	**17.58**
Deal_PermID	Unique Bloomberg identifier for the deal.	**0**
BB_Deal_Number	Bloomberg ID for the equity offering.	**0**
Issue_Date	Date the equity is publicly issued.	**0**
Offer_Year	Calendar year of the offering.	**0**
Issuer_Exchange	Exchange where issuer is listed.	**3.97**
New_Issues_Primary_Exchange	Exchange for the new security listing.	**3.98**
Issuer_Ticker	Issuer's stock ticker before offering.	**4.17**

Issuer_Name_Full	Full legal name of the issuing company.	**0**
Issuer_CUSIP6	First six characters of issuer's CUSIP.	**0**
Issuer_CUSIP8	Eight-character CUSIP for issuer/security.	**16.24**
Issuer_CUSIP9	Nine-character CUSIP for the security.	**16.24**
New_Issues_Ticker	Ticker assigned to the new security.	**4.10**
New_Issues_Company_ID	Unique ID for the issuing company.	**0.25**
New_Issues_HiTech_Group	Flags high-tech industry membership.	**0**
Not_For_Profit_Flag	True if issuer is not-for-profit.	**0**
Issue_Type	Offering type (e.g., IPO, secondary).	**0**
Regulation_Types	Regulatory framework (e.g., Reg D, Reg S).	**12.36**
VC_Backed_IPO_Issue_Flag	Flags venture capital-backed IPOs.	**21.55**
PE_Backed_IPO_Issue_Flag	Flags private equity-backed IPOs.	**0**
Marketplace	Platform where security is traded.	**0**
Market_Area	Geographical region targeted by offering.	**0**

Target_Market	Intended investor group (e.g., USA).	**0**
SIC	Standard Industrial Classification code.	**0**
Issuer_SDC_Industry	SDC-based industry classification.	**0**
Share_Type_Offered	Type of shares issued (e.g., common).	**3.47**
Security_Type_This_Mkt	Security type in the current market.	**0**
Security_Type_All_Mkts	Security type across all markets.	**0**
Offer_Price_USD	Per-share offer price in USD.	**0.05**
Free_Float_at_IPO	% of shares freely tradable at IPO.	**99.82**
Shares_Offered_All_Mkts	Total shares offered across all markets.	**0.16**
Shares_Offered_This_Mkt	Shares offered in the current market.	**0**
Primary_Amt_Offered_All_Mkts	USD value of new shares, all markets.	**14.12**
Primary_Amt_Offered_This_Mkt	USD value of new shares, current market.	**14.12**
Secondary_Amt_Offered_All_Mkts	USD value of existing shares, all markets.	**82.77**
Secondary_Amt_Offered_This_Mkt	USD value of existing shares, current market.	**82.79**

Price_at_Close_Issue_Date	Stock closing price on issue date.	**26.80**
Price_at_Close_Offer_Date	Stock closing price on offer date.	**23.18**
Price_1_Day_After	Stock price one day post-offering.	**23.34**
Price_1_Day_Before	Stock price one day pre-offering.	**23.55**
Offer_Price_Available	Flags availability of offer price data.	**0**

Table 2: Variable Definitions of external data – Industry Portfolio

Error! Bookmark not defined.

Variable Name	**Definition (As per French Fama and Kenneth R.)**
Portfolio	Industry group per Fama-French classification.
value_weighted_monthly	Monthly returns weighted by market cap.
F-F_Momentum_Factor	Return difference: high vs. low past returns.
F-F_Research_Data_5_Factors_2x3.	5-factor model returns (size, value sorted).
Mkt-RF	Market return minus risk-free rate.
SMB	Small minus big firm returns (size factor).
HML	High minus low book-to-market (value factor).

RMW	Robust minus weak profitability factor.
CMA	Conservative minus aggressive investment factor.
RF	Risk-free rate

Table 3: Variable Definitions of external data – Macroeconomic**Error! Bookmark not defined.**

Variable Name	Definition
tbill_30_return	Monthly return on 30-day U.S. T-bill.
tbond_30_return	Monthly return on 30-year U.S. T-bond.
consumer_price_index	Tracks consumer price inflation.
proj_growth_rate	Forecasted real GDP growth rate.
Disposable personal income PCT change	% change in after-tax household income.
Gross domestic product (GDP) PCT change	% change in total economic output.
Implicit regional price deflator 10 PCT change	% change in regional price deflator.
Per capita disposable personal income 7 PCT change	% change in per capita after-tax income.
Per capita personal consumption expenditures (PCE) 8 PCT change	% change in per capita spending.
Per capita personal income 6 PCT change	% change in per capita total income.
Personal consumption expenditures PCT change	% change in total consumer spending.
Personal income PCT change	% change in total household income.

Real GDP (millions of chained 2017 dollars) 1 PCT change	% change in inflation-adjusted GDP.
Real PCE (millions of constant (2017) dollars) 3 PCT change	% change in real consumer spending.
Real per capita PCE 5 PCT change	% change in real per capita spending.
Real per capita personal income 4 PCT change	% change in real per capita income.
Real personal income (millions of constant (2017) dollars) 2 PCT change	% change in real total income.
Regional price parities (RPPs) 9 PCT change	% change in regional price levels.
Total employment (number of jobs) PCT change	% change in total jobs.
Per capita personal income (dollars) 2 QoQ change	QoQ % change in per capita income.
Personal income (millions of dollars, seasonally adjusted) QoQ change	QoQ % change in total income.
Population (midperiod, persons) 1 QoQ change	QoQ % change in population.
Chain-type quantity indexes for real GDP QoQ change	QoQ % change in real GDP index.
Current-dollar GDP (millions of current dollars) QoQ change	QoQ % change in nominal GDP.
Real GDP (millions of chained 2017 dollars) 1 QoQ change	QoQ % change in real GDP.

Table 4: Variable Definitions of external data – Microeconomic**Error! Bookmark not defined.**

Variable Name	Definition
P_E_Excl_Extra	P/E ratio excluding one-time items.
P_E_Incl_Extra	P/E ratio including one-time items.
Net_Profit_Margin	% of revenue as profit after expenses.
Operating_Profit_Margin_Before_Depreciation	% of revenue as operating profit pre-depreciation.
Gross_Profit_Margin	% of revenue after cost of goods sold.
Cash_Flow_Margin	Operating cash flow to revenue ratio.
Return_on_Assets	Profit relative to total assets.
Return_on_Equity	Profit relative to shareholders' equity.
Shares_Outstanding	Total shares owned by shareholders.
Trading_Volume	Total shares traded in a period.
Current_EPS	Earnings per share for recent period.